

Task S5.01. Queries with MongoDB

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p2p review partner: Federico Urbina

We will work with a database that contains collections related to a film entertainment application:

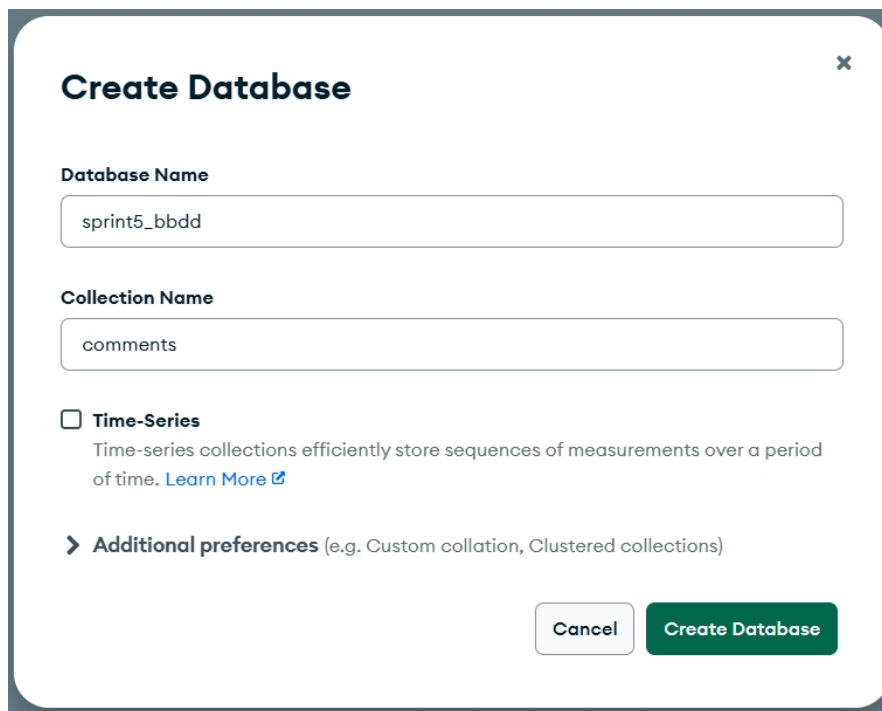
- **users:** Stores user information, including names, emails, and encrypted passwords.
- **theatres:** Contains cinema data, such as ID, location (address and geographical coordinates).
- **sessions:** Stores user sessions, including user IDs and JWT tokens for authentication.
- **movies:** Includes movie details, such as plot, genres, duration, cast, comments, release year, directors, ratings and awards.
- **comments:** Stores user comments about movies, with information about the author of the comment, movie ID, comment text, and date.

You will carry out some queries requested by the client, who is measuring whether or not you will be able to take charge of the analytical part of the project linked to their database.

Level 1

Create a database with MongoDB using the attached files as collections.

To start off, we need to create the database. Figure 1 shows how to do that, besides a collection, which is similar to MySQL tables.



The screenshot shows a 'Create Database' dialog box with the following fields and options:

- Database Name:** sprint5_bbdd
- Collection Name:** comments
- ☐ **Time-Series**
Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)
- Additional preferences** (e.g. Custom collation, Clustered collections)
- Buttons:** Cancel, Create Database

Fig 1. Creating the sprint5_bbdd database with its first table: comments

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Then we create the remaining collections in the database using the + sign at the top-right of the database at the left-side panel.

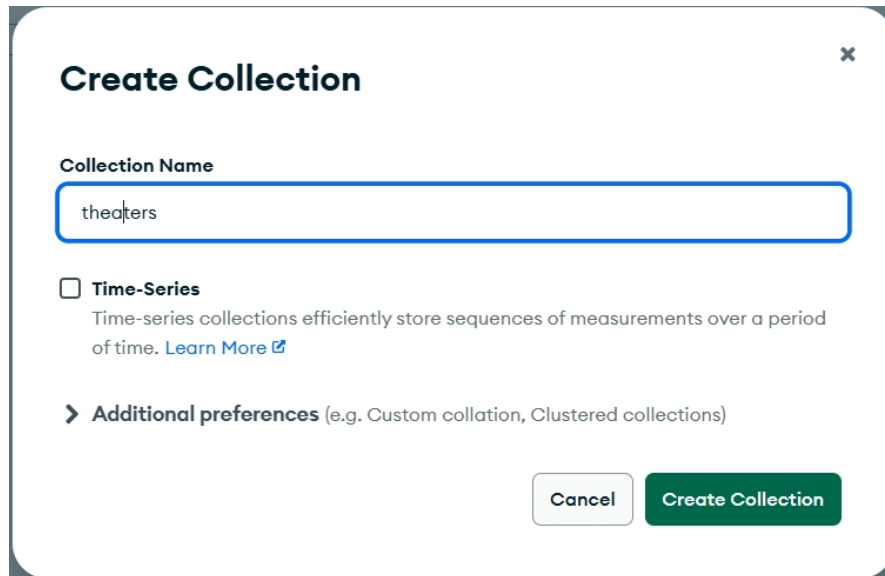


Fig 2. Creating the remaining collections

Now, by highlighting each collection, one can import the JSON file using the "Import data" button, as shown in Figures 3 and 4.

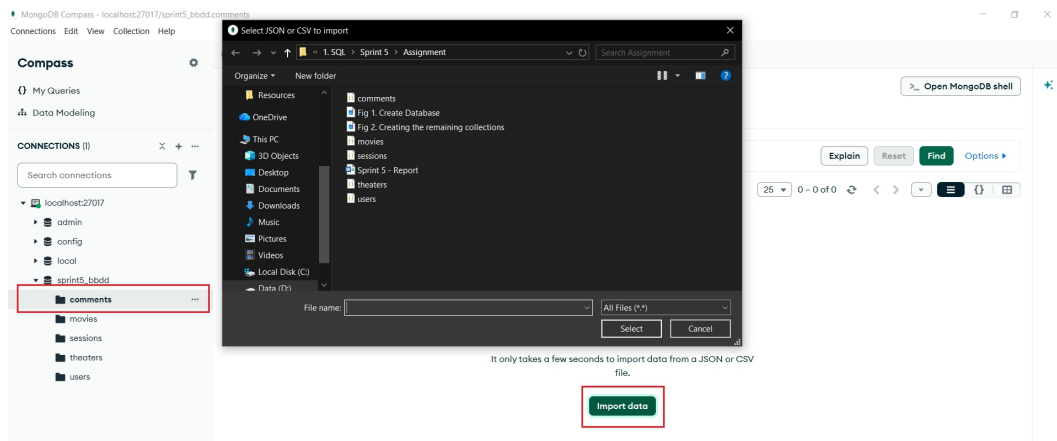


Fig 3. Importing data to each collection

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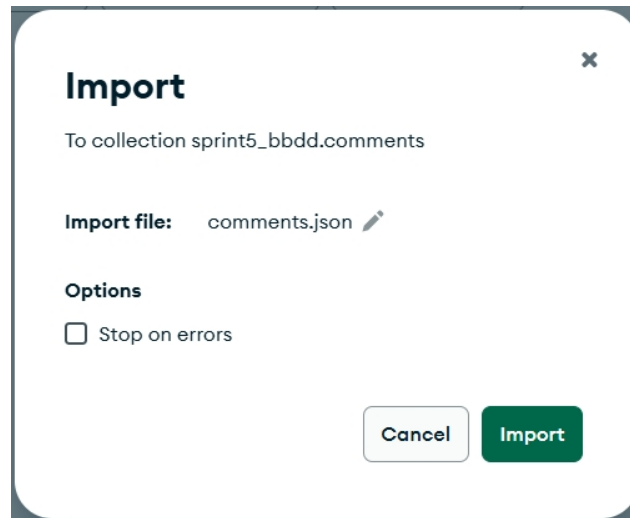


Fig 4. Importing JSON files corresponding to each collection

If all goes well, we should expect a successful import notification, as shown in Figure 5.

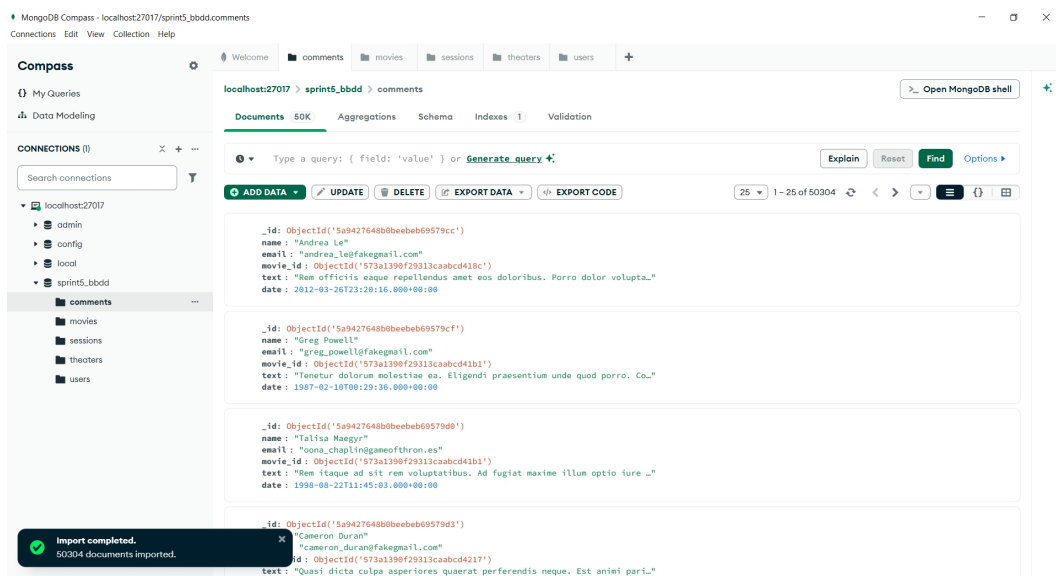


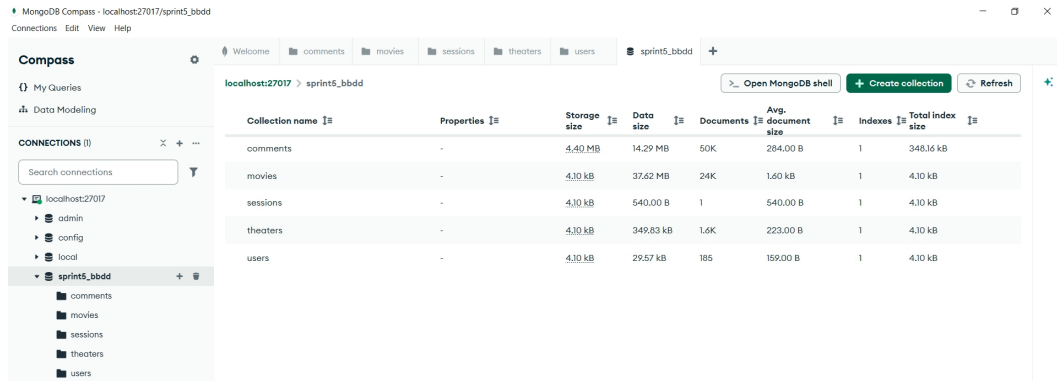
Fig 5. Successfully imported data to a collection

Eventually, we will have a database like the one shown in Figure 6.

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The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'sprint5_bbdd' database. The main panel displays a table of collections for 'localhost:27017 > sprint5_bbdd'. The table has columns for Collection name, Properties, Storage size, Data size, Documents, Avg. document size, Indexes, and Total Index size.

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total Index size
comments	-	4.40 MB	14.29 MB	50K	284.00 B	1	348.16 kB
movies	-	4.10 kB	37.62 MB	24K	1.60 kB	1	4.10 kB
sessions	-	4.10 kB	540.00 B	1	540.00 B	1	4.10 kB
theaters	-	4.10 kB	349.83 kB	1.6K	223.00 B	1	4.10 kB
users	-	4.10 kB	29.57 kB	185	159.00 B	1	4.10 kB

Fig 6. The final database is ready for the analysis

Exercise 1

- Show the first 2 comments in the database.

```
>_MONGOSH
> use sprint5_bbdd
< switched to db sprint5_bbdd
> db.comments.find().limit(2)
< [
  {
    _id: ObjectId('5a9427648b0beebe69579cc'),
    name: 'Andrea Le',
    email: 'andrea_le@fakegmail.com',
    movie_id: ObjectId('573a1390f29313caabdc418c'),
    text: 'Rem officiis eaque repellendus amet eos doloribus. Porro dolor voluptatum voluptates neque culpa molestias. Voluptate unde nulla temporibu
    date: 2012-03-26T23:20:16.000Z
  },
  {
    _id: ObjectId('5a9427648b0beebe69579cf'),
    name: 'Greg Powell',
    email: 'greg_powell@fakegmail.com',
    movie_id: ObjectId('573a1390f29313caabdc41b1'),
    text: 'Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Comodi nisi sit placeat rerum vero cupiditate neque. Dolorum nihil ve
    date: 1987-02-10T00:29:36.000Z
  }
]
```

Fig 7. The first two comments in the database

- How many users do we have registered?

```
>_MONGOSH
> db.users.countDocuments()
< 185
sprint5_bbdd>
```

Fig 8. Number of registered users

- How many movie theaters are there in the state of California?

```
>_MONGOSH
> db.theaters.countDocuments({ "location.address.state" : "CA" })
< 169
sprint5_bbdd>
```

Fig 9. Number of theaters in the state of California

- Who was the first user to register?

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```
>_MONGOSH
> db.users.find().sort({ _id: 1 }).limit(1)
< {
  _id: ObjectId('59b99db4cfa9a34dcd7885b6'),
  name: 'Ned Stark',
  email: 'sean_bean@gameofthron.es',
  password: '$2b$12$UREFwsRUoyF9CRqGNK0Lz00HM/jLhgUCNNI39RJAqMUQ74crLJ1Vu'
}
sprint5_bbdd>
```

Fig 10. The first registered user

- How many comedy films are in our database?

```
>_MONGOSH
> db.movies.countDocuments( { genres: "Comedy" } )
< 7024
sprint5_bbdd>
```

Fig 11. Number of comedy films in our database

Exercise 2

Show me all the documents for films produced in 1932, but the genre is drama or they are in French.

```
>_MONGOSH
> db.movies.find({year: 1932, $or: [{genres: "Drama"}, {languages: "French"}]}).count()
< 18
> db.movies.find({year: 1932, $or: [{genres: "Drama"}, {languages: "French"}]})
< {
  _id: ObjectId('573a1391f29313caabcd9458'),
  plot: 'A young artist draws a face at a canvas on his easel. Suddenly the mouth on the drawing comes into life and starts talking. The artist tri
runtime: 55,
rated: 'UNRATED',
cast: [
  'Enrique Rivero',
  'Elizabeth Lee Miller',
  'Pauline Carton',
  'Odette Talazac'
],
num_mflix_comments: 1,
poster: 'https://m.media-amazon.com/images/M/MV5BYWY3ODk5ZWYtYmY0ODNjA4LTk4ZWYtMzBhZDZlMjY0YTUyXkE5YkFqcGdeQXVyNzI4NDMyNTU0._V1_SY1000_SX677_AL
title: 'The Blood of a Poet',
lastupdated: '2015-09-16 13:13:05.537000000',
languages: [
  'French'
],
released: 2010-05-20T00:00:00.000Z,
directors: [
  'Jean Cocteau'
],
writers: [
  'Jean Cocteau'
],
awards: {
```

Fig 12. Showing the movie information produced in 1932 with specific conditions

We first used “count()” to know the quantity of these movies. As shown in Figure 12, 18 movies were produced in 1932 that were in French, were dramas, or were both. We used the “\$or” command to ensure the correct results.

Exercise 3

Show me all documents of American films with between 5 and 9 awards that were produced between 2012 and 2014.

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Figure 13 shows that there are 166 American movies under such conditions. To fulfill the range condition, which is equal to BETWEEN in MySQL, two operators are used: \$gte (greater than or equal) and \$lte (lower than or equal).

```
>_MONGOSH
> db.movies.find({countries: "USA", "awards.wins": {$gte: 5, $lte: 9}, year: {$gte: 2012, $lte: 2014}}).count()
< 166
> db.movies.find({countries: "USA", "awards.wins": {$gte: 5, $lte: 9}, year: {$gte: 2012, $lte: 2014}})
< [
  {
    _id: ObjectId('573a13acf29313caabd29366'),
    fullplot: "The manager of the negative assets sector of Life magazine, Walter Mitty, has been working for sixteen years for the magazine and has
    imdb: {
      rating: 7.4,
      votes: 211230,
      id: 359950
    },
    year: 2013,
    plot: 'When his job along with that of his co-worker are threatened, Walter takes action in the real world embarking on a global journey that tur
    genres: [
      'Adventure',
      'Comedy',
      'Drama'
    ],
    rated: 'PG',
    metacritic: 54,
    title: 'The Secret Life of Walter Mitty',
    lastupdated: '2015-08-31 00:10:51.747000000',
    languages: [
      'English',
      'Spanish',
      'Icelandic'
    ],
    writers: [
      'Steve Conrad (screenplay)',
```

Fig 13. American movies with specific conditions to be met

Level 2

Exercise 1

Count how many comments a user writes who uses "GAMEOFTHRON.ES" as their email domain.

```
>_MONGOSH
> db.comments.find({email: {$regex: "@gameofthron.es"}}).count()
< 22841
sprint5_bbdd>
```

Fig 14. Number of comments with a particular domain

The \$regex operator in MongoDB behaves like a wildcard in SQL. Figure 14 illustrates the simplest way to use it. We can widen our search by using the following command, although in our case, the result would be the same (since it is a full domain that comes right after @):

```
db.comments.find({email: {$regex: "/@gameofthron.es$/i"}}).count()
```

Exercise 2

How many movie theaters are there in each zip code located within the state of Washington DC (DC)?

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```
>_MONGOSH
> db.theaters.aggregate([{$match: {"location.address.state": "DC"},
  },
  {$group: {_id: "$location.address.zipcode",
    total_theaters: {$sum: 1}}
  },
  {$sort: {_id: 1}}
])
< {
  _id: '20002',
  total_theaters: 1
}
{
  _id: '20010',
  total_theaters: 1
}
{
  _id: '20016',
  total_theaters: 1
}
sprint5_bbdd>
```

Fig 15. Aggregating the results of some operators

In Figure 15, we want to aggregate the results altogether. Therefore, we used the aggregate command. We used \$match, which functions like WHERE in SQL, \$group to GROUP BY the results based on the desired criterion, \$sum to count them, and finally \$sort to sort the results.

Level 3

Exercise 1

Find all films directed by John Landis with an IMDb (Internet Movie Database) rating between 7.5 and 8.

```
>_MONGOSH
> db.movies.find({directors: "John Landis", "imdb.rating": {$gte: 7.5, $lte: 8}}).count()
< 4
> db.movies.find({directors: "John Landis", "imdb.rating": {$gte: 7.5, $lte: 8}})
< {
  _id: ObjectId('573a1397f29313caabce6d94'),
  fullplot: "Faber College has one frat house so disreputable it will take anyone. It has a second one full of white, anglo-saxon, rich young men w
  imdb: {
    rating: 7.6,
    votes: 84834,
    id: 77975
  },
  year: 1978,
  plot: 'At a 1962 college, Dean Vernon Wormer is determined to expel the entire Delta Tau Chi Fraternity, but those trouble-makers have other plan
  genres: [
    'Comedy'
  ],
  rated: 'R',
  metacritic: 82,
  title: 'Animal House',
  lastupdated: '2015-09-13 00:02:47.803000000',
  languages: [
    'English',
    'Italian'
  ],
  writers: [
    'Harold Ramis',
    'Douglas Kenney',
    'Chris Miller'
  ],
}
```

Fig 16. Movies directed by John Landis, rated between 7.5 and 8.0

There are 4 movies directed by John Landis with an IMDb rating between 7.5 and 8.0. Once again, to put a limit, we used operators \$gte and \$lte.

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Exercise 2

Show the location of all the theaters in the database on a map.

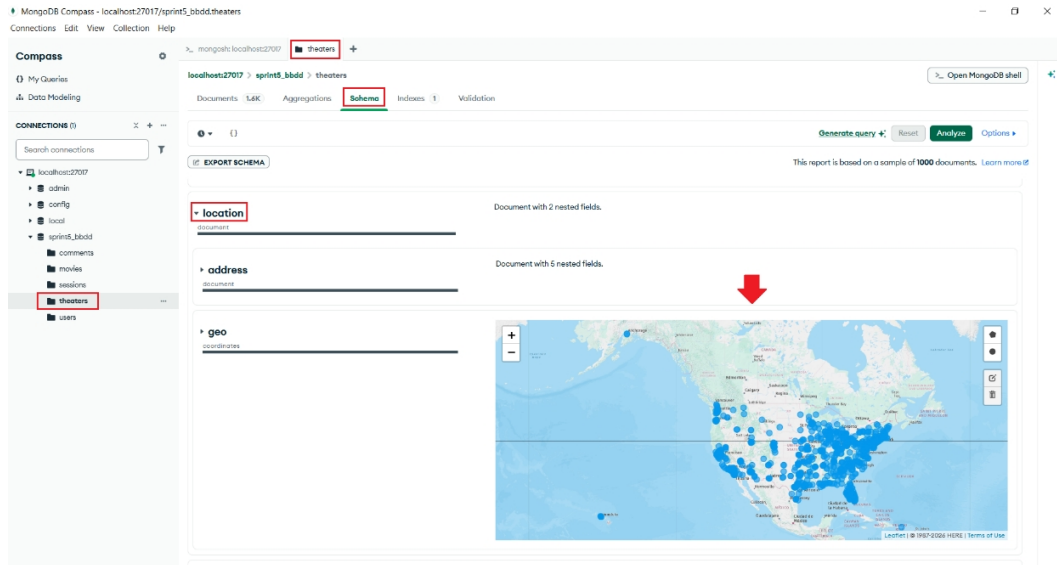


Fig 17. Location of all the theaters in the database

This exercise needs no coding! All that should be done is to click on theaters at the left panel, go to the schema tab, click Analyze schema, and then, select (drop-down) the location. The map is in the “geo” field.